## Appendix A
## Proof of Proposition 1

In this section, we prove the existence and uniqueness of NE in Game 1 separately.

### A. NE Existence in Game 1

According to Debreu-Fan-Glicksberg Theorem [22], if each decision variable is a compact and convex set, and the objective function is quasi-concave in its decision variable, the game has at least one pure NE.

In Game 1, clients choose $\alpha_i \in [0,1]$ and the server chooses $q \in [0,1]$. These are compact and convex sets.

Further, $U_i$ in Problem (3) is concave [23] in $\alpha_i$ for each client because of its second derivative is negative, as

$$\frac{\partial^2 U_i}{\partial \alpha_i^2} = -2eP_i q \leq 0. \tag{13}$$

For the server, $\phi$ is a linear function in $q$.

Therefore, we can prove that Game 1 has at least one NE.

### B. NE Uniqueness in Game 1

We start with two lemmas and then complete the proof.

**Lemma 3.** *Given a detection probability $q \in [0,1]$ of the server, the resulting attacking profile $\boldsymbol{\alpha}(q)$ is unique.*

*Proof.* For $q > 0$, we invoke Rosen's uniqueness theorem for concave games [25]. A sufficient condition for a unique joint equilibrium is that the game is Diagonally Strictly Concave (DSC), which holds if $G + G^T$ is negative definite with $G$ being the game's Jacobian matrix. In our game, the diagonal entries (self-concavity) are $G_{ii} = \frac{\partial^2 U_i}{\partial \alpha_i^2} = -2qeP_i$, and the off-diagonal entries (strategic externalities) are $G_{ij} = \frac{\partial^2 U_i}{\partial \alpha_i \partial \alpha_j} = qe\delta_i(1 - 2e\alpha_j)$ for $i \neq j$. By applying Gershgorin circle theorem to the symmetric matrix $G + G^T$, it is strictly negative definite if the following condition is satisfied:

$$|-4qeP_i| > \sum_{j \neq i} \Big| qe\delta_i(1 - 2e\alpha_j) + qe\delta_j(1 - 2e\alpha_i) \Big|. \tag{14}$$

Condition (14) reflects the inherent design of stealthy backdoor attacks [1], where the high-value gain of a successful trigger fundamentally dominates the marginal loss caused by systemic model degradation. That is, a client's own maximum backdoor-induced gain fundamentally outweighs the aggregate poisoning externalities (i.e., $2P_i \gg \sum_{j \neq i} \delta_i$). Thus, the matrix $(G + G^T)$ is strictly diagonally dominant when the practical condition (14) holds. Therefore, the game is strictly DSC, and $\boldsymbol{\alpha}(q)$ is unique. Additionally, for $q = 0$, the utility function in (3) decouples into a linear function, trivially yielding a unique boundary response for each client. □

**Lemma 4.** *Given the unique mapping $\boldsymbol{\alpha}(q)$, the server's optimal detection probability $q^*$ that forms a Nash Equilibrium is unique.*

*Proof.* Let $H(q) \triangleq y(\boldsymbol{\alpha}(q))$, where $y(\boldsymbol{\alpha})$ is the server's threshold function defined in (10). According to the server's best response in (10), the game admits three possible equilibrium regimes: $q^* = 1$ if $H(1) > 0$; $q^* = 0$ if $H(0) < 0$; or an interior equilibrium $q^* \in (0,1)$ which requires $H(q^*) = 0$.

We only need to prove that if $H(q) = 0$ occurs, the root $q^*$ is strictly unique, because $q$ is unique in the other two possible equilibriums. From Lemma 1, for a profitable client, the unconstrained optimal parameter $\theta_i(q)$ strictly decreases as $q$ increases. Because of the projection bound $[\cdot]_0^1$, the clipped optimal response $\alpha_i(q)$ is monotonically non-increasing in $q$. Furthermore, taking the partial derivative for client $k \in \mathcal{I}$ yields:

$$\frac{\partial y}{\partial \alpha_k} = 2\lambda e\alpha_k + e(S + c_k) - e\Big(\sum_{i \in \mathcal{I}} c_i\Big) \prod_{j \neq k} (1 - e\alpha_j) \geq 0. \tag{15}$$

Condition (15) reveals a practical parameter setting where the punishment $S$ and poisoning deficit $\lambda$ dominate the unlearning compensation costs. Thus, the strictly positive derivative in Condition (15) ensures $y(\boldsymbol{\alpha})$ is strictly monotonically increasing with respect to $\boldsymbol{\alpha}$. Consequently, the composite function $H(q)$ is monotonically non-increasing in $q$.

The function $H(q)$ can only possess multiple roots if it evaluates to zero and remains constant (flat) over an interval of $q$. $H(q)$ is only constant if the profile $\boldsymbol{\alpha}(q)$ is invariant, which requires all clients to be fully projected to the boundaries (0 or 1). However, if all clients are fully deterred such that $\boldsymbol{\alpha} = \mathbf{0}$, then $H(q) = y(\mathbf{0}) = -\sigma < 0$. Conversely, consider $\boldsymbol{\alpha} = \mathbf{1}$, we have $H(q) = y(\mathbf{1}) = \lambda eI + e\sum_{i \in \mathcal{I}}(S + c_i) - \sigma + (\sum_{i \in \mathcal{I}} c_i)(1 - e)^I - \sum_{i \in \mathcal{I}} c_i$. We require the system to have a practical setting $y(\mathbf{1}) > 0$ to ensure that, at least in the worst-case scenario of a maximum, all-client attack ($\alpha_i = 1, \forall i$), the server's economic benefits from preventing catastrophic model degradation and collecting punishments strictly outweigh its operational detection and unlearning costs. Therefore, the continuous function $H(q)$ transitions from strictly negative to positive on the two sides, and its root lies in $(0,1)$. For $H(q)$ to equal zero, there must be at least one active client $k$ strictly in the interior $\alpha_k \in (0,1)$, where $\alpha_k(q)$ strictly decreases in $q$. Thus, $H(q)$ strictly decreases exactly when it crosses zero, guaranteeing at most one unique root $q^*$. □

Finally, we complete the proof for NE uniqueness in Proposition 1. By Lemma 3, every $q$ maps to a strictly unique joint client best response $\boldsymbol{\alpha}(q)$. By Lemma 4, the server's best response to this monotonic mapping yields exactly one stable detection probability $q^*$. Therefore, the combined strategy profile $(q^*, \boldsymbol{\alpha}(q^*))$ constitutes the unique Nash Equilibrium of Game 1.

## Appendix B
## Proof of Proposition 3

We first prove that (11) is a NE because all players in Game 1 perform their best response, given the other player's strategies.

For the server, when $\alpha_k^* = \frac{-e(S-\sum_{j\neq k} c_j)+\sqrt{e^2(S-\sum_{j\neq k} c_j)^2+4\lambda e\sigma}}{2\lambda e}$ and $\alpha_j^* = 0, \forall j \neq k$, its threshold function in (10) is

$$y(\alpha_k^*, \boldsymbol{\alpha}_{-k}^*) = \lambda e \alpha_k^{*2} + e(S - \sum_{j\neq k} c_j)\alpha_k - \sigma = 0. \tag{16}$$

Thus, choosing $q^* \in (0,1)$ in (11) is its best response.

For client $k$, applying $q^*$ and $\alpha_j^* = 0$, we can compute

$$\theta_k(q^*, \boldsymbol{\alpha}_{-k}^*) > 0. \tag{17}$$

Thus, $\alpha_k^* > 0$ holds.

For client $j$, applying $q^*$ and $\alpha_k^*$, we can compute

$$\theta_j(q^*, \boldsymbol{\alpha}_{-j}^*) = \frac{S+\beta_k M}{2P_j}\frac{P_j-A_j}{P_k-A_k} + \frac{\delta_j(1-e\alpha_k^*)\alpha_k^*-S-\beta_j M}{2P_j} < 0. \tag{18}$$

Thus, choosing $\alpha_j^* = 0$ is its best response.

## Appendix C
## Proof of Corollary 1

To prove (12), we first compute the gap between our mechanism and No-Attack in the server's payoff and clients' total utility.

### A. Server's Payoff Gap

Applying the equilibrium in (11) to server's payoff in (5), we can get

$$\begin{aligned}\phi_{Ours} &= \lambda e q^* \alpha_k^{*2} + eq(S - \textstyle\sum_{j\neq k} c_j)\alpha_k^* - \sigma q^* - \lambda\alpha_k^* \\ &= q^* y(\boldsymbol{\alpha}^*) - \lambda\alpha_k^* \\ &= -\lambda\alpha_k^*,\end{aligned} \tag{19}$$

where the second to the third line is because of (16).

Also, we can find that $\phi_{No-Attack} = 0$ because $\alpha_i = 0, \forall i \in \mathcal{I}$ and $q = 0$ in No-Attack baseline.

### B. Clients' Total Utility Gap

Applying the equilibrium in (11), we can derive the client's utility in our mechanism as

$$U_k = -eP_k q^* \alpha_k^{*2} + (P_k - A_k - eq^*(S + \beta_k M))\alpha_k^* + \beta_k M, \tag{20}$$

and

$$U_j = e\delta_j q^* \alpha_k^{*2} - \delta_j \alpha_k^* + \beta_j M, \forall j \neq k. \tag{21}$$

Also, we can find $U_i = \beta_i M, \forall i \in \mathcal{I}$ in No-Attack baseline.

### C. Social Welfare Gap

Summing up the server's payoff in (19), client's utility in (20) and (21), the social welfare in our mechanism is

$$\begin{aligned}SW_{Ours} &= eq^*(\textstyle\sum_{j\neq k}\delta_j - P_k)\alpha_k^{*2} \\ &\quad +(P_k - \lambda - A_k - eq^*\beta_k M - \textstyle\sum_{j\neq k}\delta_j - eSq^*)\alpha_k^* \\ &\quad +\textstyle\sum_{i\in\mathcal{I}}\beta_i M.\end{aligned} \tag{22}$$

For No-Attack baseline, applying $q = 0$ and $\alpha_i = 0, \forall i \in \mathcal{I}$, we can find $SW_{No-Attack} = \sum_{i\in\mathcal{I}} \beta_i M$.

Thus, the social welfare difference between our mechanism and No-Attack baseline is

$$\begin{aligned}&SW_{No-Attack} - SW_{Ours} \\ &= -eq^*(\textstyle\sum_{j\neq k}\delta_j - P_k)\alpha_k^{*2} \\ &\quad -(P_k - \lambda - A_k - eq^*\beta_k M - \textstyle\sum_{j\neq k}\delta_j - eSq^*)\alpha_k^*,\end{aligned} \tag{23}$$

which is a quadratic function in $\alpha_k^*$ and approaches zero because $\lim_{S\to\infty}\alpha_k^* = 0$.